

Gravitational Wave Echoes from the Algebraic Domain Wall: The Kerr-Sedenion Model of Black Hole Interiors

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Abstract

We propose a resolution to the Black Hole Information Paradox based on the Axiomatic Physical Homeostasis (APH) framework, treating gravitational collapse not as a singularity formation, but as a vacuum phase transition. We posit that when curvature invariants exceed a critical threshold, the vacuum geometry reverts from the sector of the Albert Algebra ($J(3, \mathbb{O})$) to a Sedenionic sector ($\mathbb{S}$) via homeostatic phase transitions. Consequently, the black hole interior is a finite-volume condensate—a “Sedenion Star”—governed by a Super-Linear Stiffness $\Gamma_{\mathbb{S}} \approx \pi$. This work supersedes previous APH treatments by explicitly extending the model to the rotating Kerr metric regime. We identify the Event Horizon as the topological domain wall separating the exterior vacuum from the algebraic bulk. Using the Thin Filament Code (TFC) solver, we model the ringdown of this rotating stiff core and predict the existence of **Gravitational Wave Echoes** with a reflection coefficient of $R \approx 1.9\%$. We derive the characteristic time delay and frequency modulation of these echoes as a function of the spin parameter a , providing a falsifiable signature for LIGO/Virgo data analysis.

1 Introduction: The History of the Paradox

Since the discovery of the Schwarzschild solution in 1916, the physical nature of the gravitational singularity has remained the most vexing problem in theoretical physics. The Information Paradox, formalized by Hawking (1976), highlights a fundamental conflict between General Relativity (GR) and Quantum Mechanics (QM): GR predicts the destruction of information at the singularity, while QM demands unitary time evolution.

Several candidates have been proposed to resolve this, including:

- **Fuzzballs (Mathur):** Replacing the horizon with a string theory microstate geometry.
- **Firewalls (AMPS):** Positing a high-energy curtain of quanta at the horizon to preserve unitarity.
- **Gravastars (Mazur & Mottola):** Replacing the interior with a de Sitter vacuum core.

While these models rely on quantum field theoretic corrections, the Axiomatic Physical Homeostasis (APH) framework proposes a geometric-algebraic solution. We argue that the collapse is halted not by quantum pressure, but by **Algebraic Filtration**. The singularity is prevented because the vacuum possesses a “Sedenionic Resistance”—a fundamental stiffness arising from the saturation of the non-associative geometry.

2 The Phase Transition Hypothesis

In the APH cosmological model, the Big Bang is defined as the filtration of Sedenionic zero divisors ($\mathbb{S} \rightarrow \mathbb{O}$) to establish a consistent spatial metric. We propose that gravitational collapse is the inverse process: the **Re-Melting** of the vacuum.

2.1 The Associator Hazard Limit

In the exterior universe, physics is constrained to the Octonions ($\mathbb{O}$) to satisfy the Axiom of Observability. However, as the Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ diverges during collapse, the energy density enables the local excitation of higher geometric dimensions ($n \geq 16$). The vacuum undergoes a local phase transition $\mathbb{O} \rightarrow \mathbb{S}$. The resulting object is a material body composed of high-dimensional algebraic foam, trapped within our spacetime.

2.2 Equation of State: The π -Limit

To define the physical properties of this core, we must derive its Equation of State (EOS). In the Thin Filament Code (TFC) formalism [Schutza, 2025], the stiffness of the vacuum is parameterized by the effective polytropic index Γ .

- **Exterior ($\mathbb{O}$):** $\Gamma_{\mathbb{O}} \approx 2.91$.
- **Interior ($\mathbb{S}$):** We postulate that the Sedenion condensate saturates the geometric limit of the flux tube topology. The maximum “hoop stress” a geometry can support before the ratio of circumference to diameter fails is defined by π .

Thus, the Sedenion Core is a “Super-Stiff” fluid with:

$$P = K\rho^\pi, \quad \Gamma_{\mathbb{S}} \equiv \pi \approx 3.14159 \quad (1)$$

This stiffness provides the pressure support necessary to halt collapse at a finite radius, preventing the formation of a mathematical singularity.

3 The Kerr-Sedenion Metric

astrophysical black holes possess angular momentum J . We must therefore extend the APH model to the Kerr regime. The standard Kerr metric describes a vacuum solution ($T_{\mu\nu} = 0$). In the APH model, this solution holds only for $r > r_{wall}$. Inside the domain wall, the metric is that of a rotating fluid sphere with stiffness $\Gamma = \pi$.

3.1 The Algebraic Ergosphere

In the Kerr metric, the ergosphere is the region where frame-dragging forces co-rotation. In the APH model, the interaction between frame-dragging and the Sedenion zero-divisors is critical. Let $\mathcal{Z}(\mathbb{S})$ be the manifold of zero divisors isometric to G_2 [Reggiani, 2024]. The angular momentum vector J couples to the G_2 structure constants f_{ijk} .

$$\mathcal{L}_{int} \propto J^i \epsilon_{ijk} \langle \mathcal{Z}^{jk} \rangle \quad (2)$$

Because the interior preserves **Power Associativity** ($x^n x^m = x^{n+m}$), the rotation generator $e^{iJ\phi}$ remains well-defined. The Sedenion core can rotate, but it deforms into an oblate spheroid.

3.2 The Domain Wall Location

We identify the Event Horizon r_+ not as a causal boundary of no return, but as the **Phase Boundary** (Domain Wall) between the Octonionic and Sedenionic phases.

$$r_{wall} \approx r_+ + \epsilon \ell_P \quad (3)$$

where ϵ is a small parameter related to the healing length of the algebraic transition.

4 Gravitational Wave Echoes

When a binary black hole system merges, the resulting ringdown is traditionally modeled as the relaxation of the perturbed photon sphere. In the APH model, the ringdown includes a secondary phenomenon: waves trapped between the angular momentum barrier and the stiff Sedenion core.

4.1 Impedance Mismatch and Reflection

Gravitational waves propagating across the domain wall experience a change in the effective refractive index of spacetime. The acoustic impedance Z of the vacuum scales with the square root of the stiffness: $Z \propto \sqrt{\Gamma}$. The Reflection Coefficient R at the domain wall is given by the Fresnel equations for the vacuum impedance:

$$R = \left(\frac{\sqrt{\Gamma_S} - \sqrt{\Gamma_O}}{\sqrt{\Gamma_S} + \sqrt{\Gamma_O}} \right)^2 \approx \left(\frac{\sqrt{3.1416} - \sqrt{2.91}}{\sqrt{3.1416} + \sqrt{2.91}} \right)^2 \quad (4)$$

$$R \approx \left(\frac{1.772 - 1.706}{3.478} \right)^2 \approx (0.019)^2 \approx 3.6 \times 10^{-4} \quad (\text{Power}) \quad (5)$$

In terms of amplitude (strain h), the reflection is linear:

$$\mathcal{R}_{amp} \approx 0.019 \quad (6)$$

This implies that roughly $\sim 2\%$ of the incident wave amplitude is reflected back into the cavity.

4.2 Echo Timing in the Kerr Regime

The time delay between the main merger signal and the first echo depends on the spin of the black hole $a = J/M$. For a wave bouncing between the angular momentum barrier (r_{peak}) and the domain wall ($r_{wall} \approx r_+$), the delay is:

$$\Delta t_{echo} \approx \frac{2r_+}{c} \int_{r_+}^{r_{peak}} \frac{dr}{1 - \frac{2M}{r} + \frac{a^2}{r^2}} \quad (7)$$

Due to the coordinate singularity, this integral diverges logarithmically in GR. However, in APH, the reflection occurs at the physical surface r_{wall} . The characteristic time scale is:

$$\Delta t_{echo} \approx \frac{2GM}{c^3} (1 + (1 - a^2)^{-1/2}) \times \ln \left(\frac{r_{wall}}{r_+ - r_{wall}} \right) \quad (8)$$

Prediction: Highly spinning black holes ($a \rightarrow 1$) will exhibit distinct echo timings compared to Schwarzschild ($a = 0$) predictions. The “ringing” of the Sedenion core is modulated by the frame-dragging frequency Ω_H .

5 Conclusion

The Black Hole is not a hole; it is a **Sedenion Star**. By explicitly modeling the interior as a π -stiff algebraic condensate within the Kerr metric, we provide a self-contained resolution to the singularity problem that does not rely on ad-hoc quantum corrections. The predicted gravitational wave echoes, with a reflection amplitude of $\approx 1.9\%$, offer a direct observational test of the Sedenion hypothesis using existing LIGO/Virgo data.